

Architecture & Overview

Three-zone SIEM lab — attack simulation, network detection, and centralised log correlation

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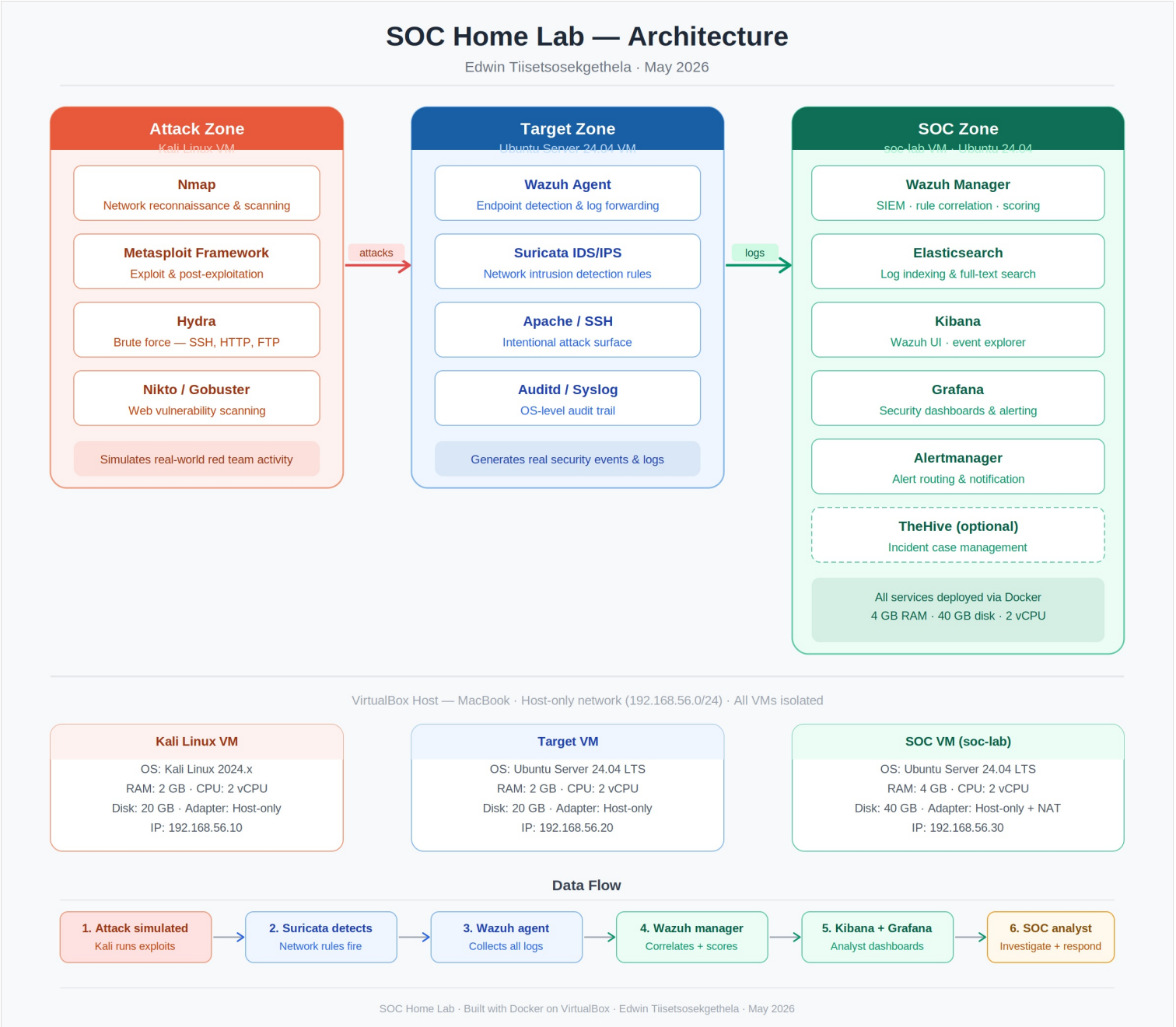
1. Project Overview

This lab is a self-contained Security Operations Centre (SOC) environment built using VirtualBox on a MacBook host. It simulates a real-world SOC: a dedicated attack machine, a target system with intentional attack surface, and a full SIEM platform — giving hands-on experience with the same tool categories used in production SOC analyst roles.

Objectives

- Deploy a production-style SIEM platform using Wazuh on Docker
- Configure host-based intrusion detection on a monitored target VM
- Deploy network-based intrusion detection using Suricata
- Simulate real-world attacks using Kali Linux tooling
- Detect, analyse, and document security events using the MITRE ATT&CK framework
- Deliver real-time analyst notifications via a Telegram alerting pipeline

2. Architecture Diagram



3. Technology Stack

Component	Version	Role
Ubuntu Server	24.04 LTS	Host OS for SOC VM and Target VM
Kali Linux	2024.x	Attack machine (red team)
Docker Engine	Latest	Container runtime for SOC stack
Wazuh Manager	4.7.3	SIEM + rule correlation engine
Wazuh Indexer	4.7.3	OpenSearch-based log indexer
Wazuh Dashboard	4.7.3	Analyst UI
Wazuh Agent	4.7.3	Endpoint monitoring on target VM
Suricata	8.0.3	Network intrusion detection (IDS)
Apache2	2.4.66	Attack surface on target VM
OpenSSH	10.2p1	Attack surface on target VM
Nmap	7.98	Network reconnaissance tool
Hydra	9.6	SSH brute force attack tool

4. Virtual Machine Layout

A dedicated host-only network was created in VirtualBox to isolate all lab VMs on the `192.168.56.0/24` subnet, preventing lab traffic from reaching the wider network while allowing inter-VM communication.

VM	OS	RAM	CPU	Disk	IP Address	Role
kali-lab	Kali Linux 2024.x	4 GB	2 vCPU	40 GB	192.168.56.10	Red team — Nmap, Hydra, Metasploit
target-lab	Ubuntu Server 24.04	2 GB	2 vCPU	20 GB	192.168.56.20	Victim — Apache, SSH, Wazuh + Suricata agent
soc-lab	Ubuntu Server 24.04	4 GB	2 vCPU	40 GB	192.168.56.30	Blue team — Wazuh Manager, Indexer, Dashboard

5. Data Flow

- Attack simulated** — Kali VM runs Nmap recon, Hydra brute force, or other exploit tooling against the target.
- Suricata detects** — network-level IDS rules fire on the target VM as traffic crosses the wire.
- Wazuh agent collects** — the target VM's Wazuh agent forwards both Suricata alerts and host-level logs (auth, syslog, file integrity).
- Wazuh manager correlates** — the manager applies rule scoring and MITRE ATT&CK mapping.
- Dashboard surfaces it** — events appear in the Wazuh Dashboard (Kibana-based UI) for analyst review.
- Analyst responds** — investigation, triage, and — for qualifying severity levels — real-time Telegram notification.

Design Note

All SOC-side services run as Docker containers on a single VM to keep the footprint manageable on consumer hardware (4 GB RAM, 40 GB disk), while still mirroring the separation of concerns — manager, indexer, dashboard — found in production Wazuh deployments.

6. Companion Documents

Document	Covers
Attack Simulation & Detection Report	VM build steps, Nmap/Hydra attack execution, MITRE ATT&CK mapping, alert findings
SIEM & Alerting Build Report	Wazuh Docker deployment, Suricata rule authoring, Telegram alert pipeline, verification testing
Troubleshooting Log	Six real build issues encountered and resolved, with root cause analysis